

Dr. Yinglong Dai

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Research Interests

Interpretable deep learning, multimodal deep learning, hierarchical deep reinforcement learning, intelligent healthcare process

Education

Ph.D. degree in Computer Science,
Central South University, China, 2018, Advisor: Guojun Wang

M.S. degree in Control Theory & Control Engineering,
Northeastern University, China, 2012, Advisor: Xianwen Gao

B.S. degree in Automation,
Northeastern University, China, 2010



Research and Work Experience

Associate Professor, College of Information Science and Engineering, Hunan Normal University, 2018-Now

Visiting Scholar, Department of Computer and Information Systems, University of Aizu, Japan, 2024-2025

Electronic Engineer, Research Institute of Intelligent Engineering, SANY Heavy Industry Co., Ltd. 2012-2013

Selected Publications (https://www.researchgate.net/profile/Yinglong_Dai)

- [1] Yinglong Dai, Zhi Yi, Qiangfu Zhao, Ming Chen, and Guojun Wang, Framework for Hierarchical Deep Reinforcement Learning with Conceptual Embedding. *Neural Networks*, 2026, 199: 108769.
- [2] Yinglong Dai, Zheng Yan, Jiangchang Cheng, Xiaojun Duan, and Guojun Wang, Analysis of Multimodal Data Fusion from an Information Theory Perspective. *Information Sciences*, 2023, 623: 164-183.
- [3] Yinglong Dai, Haibin Ouyang, Hong Zheng, Han Long, and Xiaojun Duan. Interpreting a Deep Reinforcement Learning Model with Conceptual Embedding and Performance Analysis. *Applied Intelligence*, 2023, 53: 6936–6952.
- [4] Yinglong Dai, Guojun Wang, Khan Muhammad, and Shuai Liu, A Closed-Loop Healthcare Processing Approach based on Deep Reinforcement Learning. *Multimedia Tools and Applications*, 2022, 81(3): 3107–3129.
- [5] Yinglong Dai, Guojun Wang. A Deep Inference Learning Framework for Healthcare. *Pattern Recognition Letters*, 2020, 139: 17-25.
- [6] Yinglong Dai, Guojun Wang, Jianhua Dai, and Oana GEMAN, A Multimodal Deep Architecture for Traditional Chinese Medicine Diagnosis. *Concurrency and Computation: Practice and Experience*, 2020, 32(19): e5781.

Projects

- [1] National Natural Science Foundation of China (No. 62306110), Research on an Intelligent Framework of Healthcare Process Based on Information Theory, Jan. 2024 – Dec. 2026.
- [2] Hunan Provincial Natural Science Foundation (No. 2020JJ5367), Research of Information Fusion in Multimodal Deep Learning Models, Jan. 2020 – Dec. 2022.

Honor

National College Students Electronic Design Competition, Sound Guided Smart Car, National Second Prize, 2009.